Commodore 128 — Video & Custom ICs

8563 VDC · 8722 MMU · 8721 PLA · 8564 VIC-IIE · ALL DATASHEET-VERIFIED

DIP pinouts from the top, notch up, pin 1 top-left. Power **red**, ground grey, clocks blue. Verify against the datasheet before powering up.

Legend — reading the pinouts

notation

`/NAME` **Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. `/RES /IRQ /CS /WE`.

`NAME` **Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. `R/W` is a level: HIGH = read, LOW = write.

Colour key: Vcc / Vdd = power · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

8563 VDC 80-col video

48-DIP

CCLK	1	48	/CAS
DCLK	2	47	/RAS
HSYNC	3	46	R
CS	4	45	G
NC	5	44	B
NC	6	43	I
/CS	7	42	DD7
/RS	8	41	DD6
R/W	9	40	DD5
D7	10	39	DD4
D6	11	38	DD3
VSS	12	37	VDD
D5	13	36	DD2
D4	14	35	DD1
D3	15	34	DD0
D2	16	33	DA7
D1	17	32	DA6
D0	18	31	DA5
DISPEN	19	30	DA4
VSYNC	20	29	DA3
DR/W	21	28	DA2
INIT	22	27	DA1
/RES	23	26	DA0
TEST	24	25	LPEN

8563 Video Display Controller: 80-column RGBI with its own 16/64KB video RAM (DA0-7 + DD0-7). Note **two** chip-selects: pin 4 = CS (active high), pin 7 = /CS (active low); pins 5/6 are NC. Runs hot, low yield, a known failure. **Fails:** no/garbled 80-col.

8722 MMU memory mgmt

48-DIP

VDD	1	48	SENSE40
/RESET	2	47	128/64
TA15	3	46	/EXROM
TA14	4	45	/GAME
TA13	5	44	FSDIR
TA12	6	43	/Z80EN
TA11	7	42	D7
TA10	8	41	D6
TA9	9	40	D5
TA8	10	39	D4
/CAS1	11	38	D3
/CAS0	12	37	D2
I/O SEL	13	36	D1
MS1	14	35	D0
MS0	15	34	VSS
AEC	16	33	PHI0
MUX	17	32	R/W
A0	18	31	A15
A1	19	30	A14
A2	20	29	A13
A3	21	28	A12
A4/5	22	27	A11
A6/7	23	26	A10
A8	24	25	A9

8722 Memory Management Unit: selects 8502/Z80, banks ROM/RAM, sets the common areas and zero-page/stack relocation, drives the translated address TA8-TA15. Pins 22/23 are multiplexed (A4/5, A6/7). **Fails:** dead in 64 or 128 mode, bad banking.

8721 PLA decode

48-DIP

A15	1	48	VCC
A14	2	47	CLK
A13	3	46	/CHAROM
A12	4	45	/COLRAM
A11	5	44	GWE
A10	6	43	/IOACC
VICFIX	7	42	/VIC
DMAACK	8	41	CASENB
AEC	9	40	DWE
R/W	10	39	DIR
GAME	11	38	/IOCS
EXROM	12	37	/ROM1
Z80 EN	13	36	/ROM2
Z80 I/O	14	35	/ROM3
64/128	15	34	/ROM4
I/O SE	16	33	/FROM
ROMBNKHI	17	32	CLRBNK
ROMBNKLO	18	31	/ROMH
VMA4	19	30	/ROML
VMA5	20	29	/SDEN
BA	21	28	N/C
LORAM	22	27	128/256
HIRAM	23	26	VA14
VSS	24	25	CHAREN

8721 PLA (not the C64's 82S100): address decode, ROM/RAM bank and chip-selects (/ROM 1-4, /CHAROM, /COLRAM, /FROM, /IOCS...). LORAM/HIRAM come from the 8502 port. **Fails:** dead machine, wrong banking, no I/O.

8564/8566 VIC-IIe 40-col video

48-DIP

DB6	1	48	VCC
DB5	2	47	DB7
DB4	3	46	DB8
DB3	4	45	DB9
DB2	5	44	DB10
DB1	6	43	DB11
DB0	7	42	A10
/IRQ	8	41	A9
/LP	9	40	A8
BA	10	39	A7
/DMAR	11	38	A6
AEC	12	37	A5
/CS	13	36	A4
R/W	14	35	A3
/DMAA	15	34	A2
CL	16	33	A1
S/LUM	17	32	A0
1MHz	18	31	A11
/RAS	19	30	PHIN
CAS	20	29	PHCL
MU	21	28	K2
/IDACC	22	27	K1
2MHz	23	26	K0
VSS	24	25	Z80PHI

8564 (NTSC) / 8566 (PAL) VIC-IIe: C64-compatible 40-column composite video, plus the three extra keyboard-scan lines (K0-K2) and the Z80PHI / 1MHz / 2MHz clock lines the C128 needs. Pinout verified from the 8566 datasheet. **Fails:** no/garbled 40-col display.

Test & Repair

tips

`No 40-col`	VIC-IIe / VIC RAM
`No 80-col`	8563 VDC (runs hot)
`Garbled banks`	8722 MMU / 8721 PLA
`VDC RAM`	4416 → 4464 for 64K upgrade

The 8563 VDC and the MMU/PLA are the distinctive C128 failure points.

Verification notes

⚠ verify these

`All four chips now`	datasheet-verified — ready to graduate
`8563 VDC`	note dual CS: pin 4 (high) + pin 7 (low)
`8722 MMU`	pins 22/23 are multiplexed (A4/5, A6/7)

Amber pins and **⚠ flagged chips** are my best reconstruction from incomplete/garbled sources — confirm each against a datasheet before use.